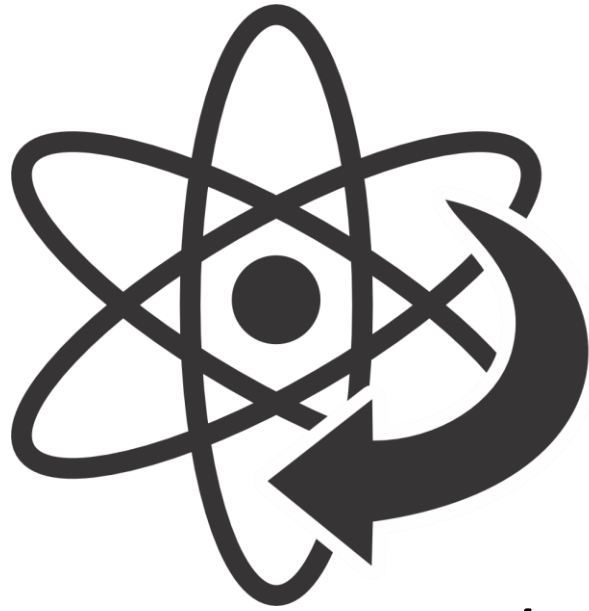




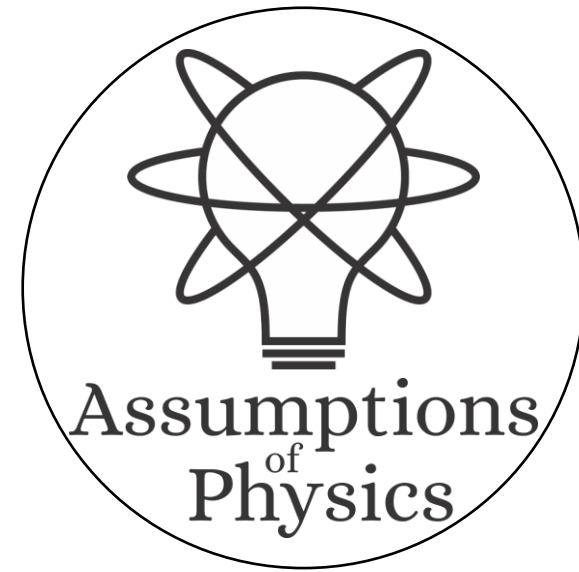
Reverse Physics

turning physics inside-out



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Physics Department
University of Michigan

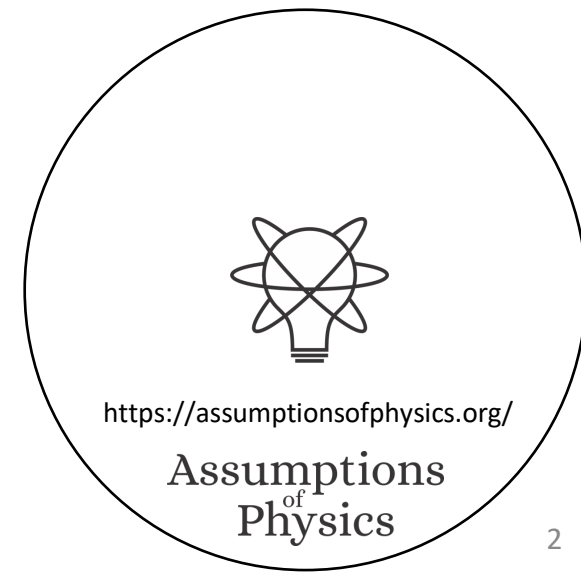


The foundations of physics has a methodology problem

Reverse Physics aims to adopt approaches that are more in line with foundations of mathematics, to break physics into a set of physical and mathematical conditions with well-understood logical relationships

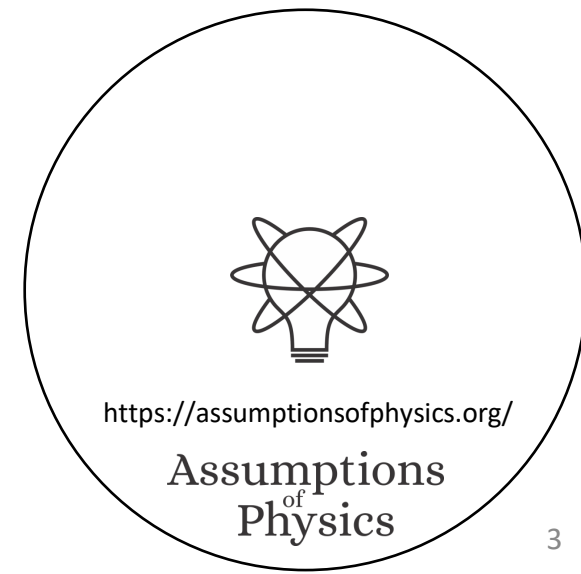


*“When the math is derived from the right physical assumptions,
the physical assumptions can be derived from the math.”*



Goals of Reverse Physics

- Framework/interpretation/formulation/theory neutral
 - Allow for each result to be independently formulated from the other
- Uncover “paradoxes” and develop standards of rigor appropriate for physics
- Clarify the logical dependencies of physical/mathematical conditions
- Identify core constitutive requirements of physical theories
 - The “big four” of Reverse Physics

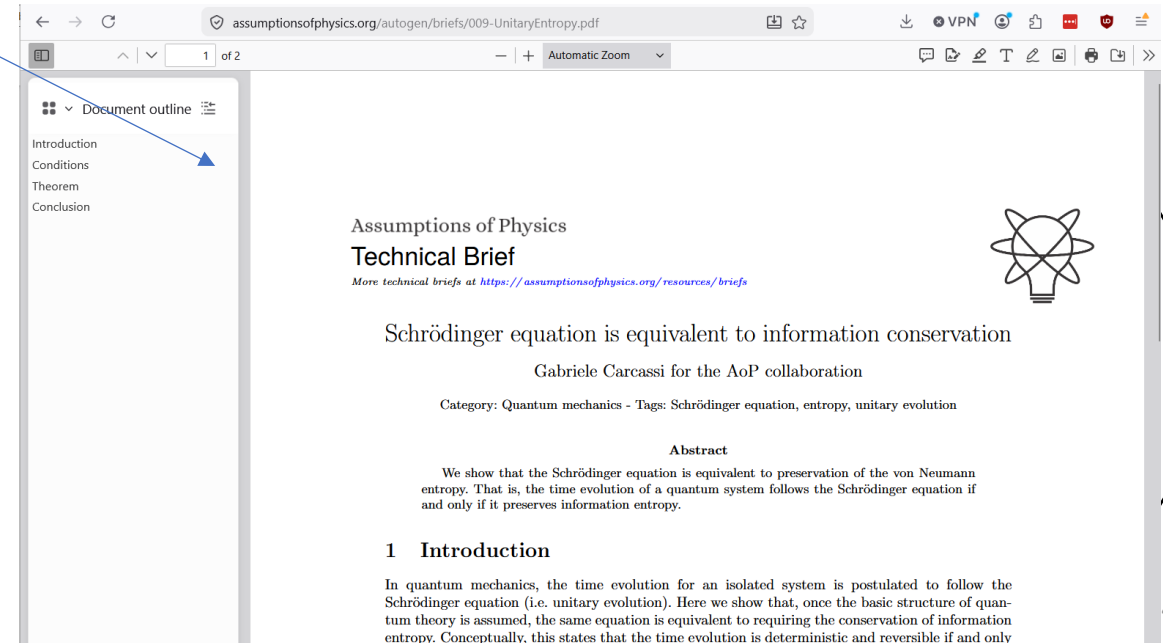


Show, don't tell!

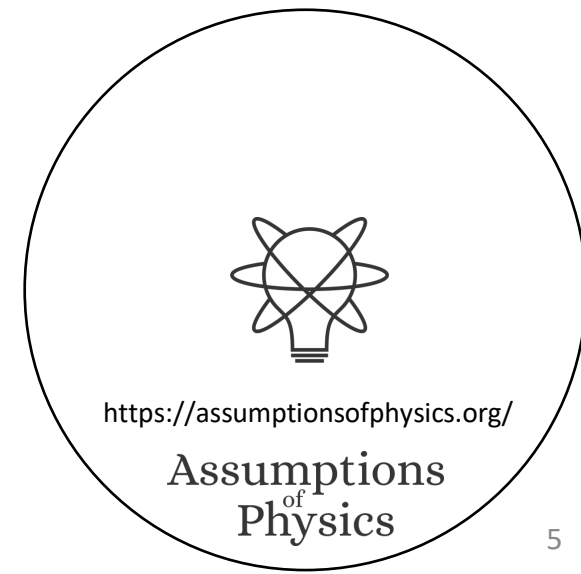
And now a series of random results to show how Reverse Physics works

Warning: no time to go through each in details

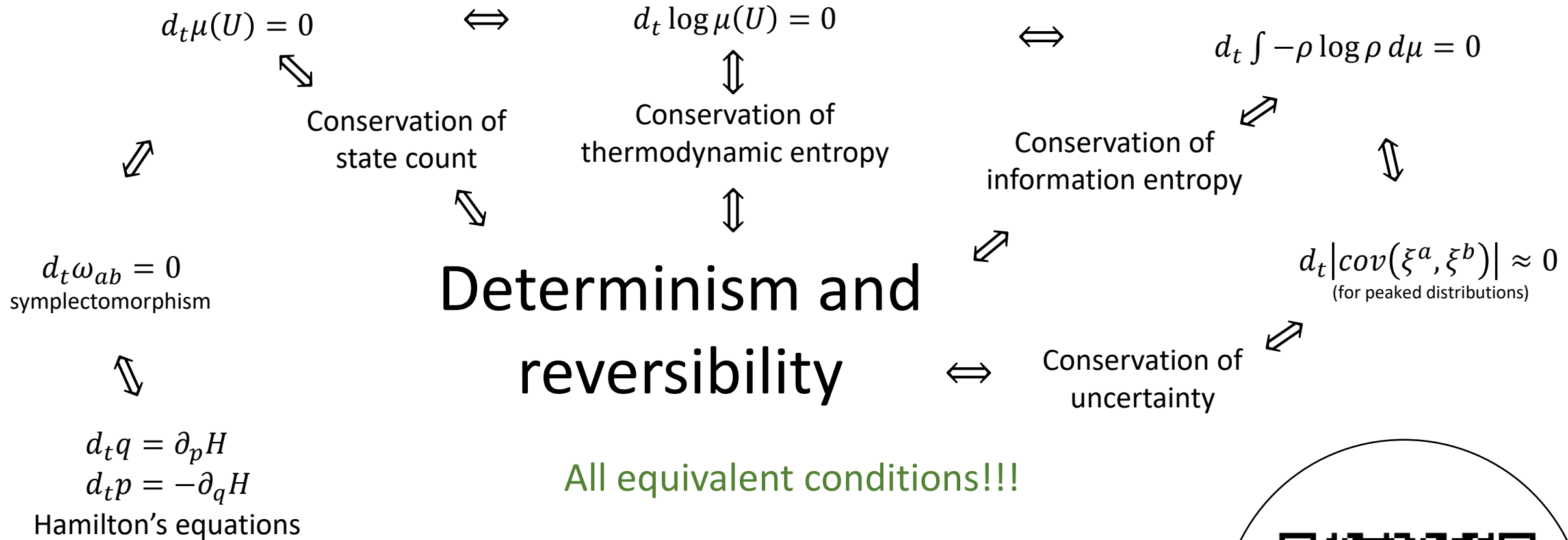
Wherever possible, QRCode to papers or technical briefs



The actual content of mathematical structures



Hamiltonian mechanics 1 DOF $\iff$ Det/Rev

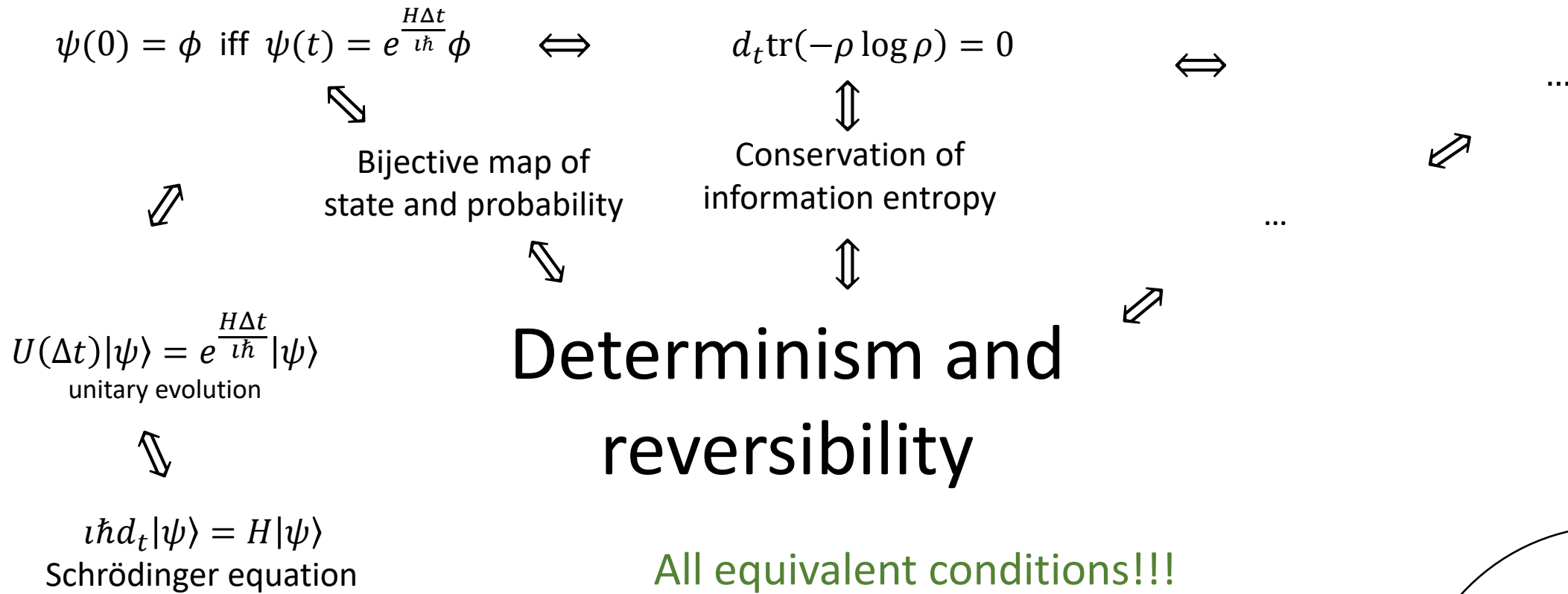


Base theory: State: $\xi^a = [q \ p] \in \mathbb{R}^2$

Count of states: $\mu(U) = \int_U dq dp = \int_U \omega$



Schrödinger equation $\Leftrightarrow$ Det/Rev



Base theory: State: $\psi \in P(\mathcal{H})$

Probability of ψ given ϕ : $p(\psi|\phi) = |\langle\psi|\phi\rangle|^2$



Reversing the principle of stationary action

DR

$$\nabla \cdot \vec{S} = 0$$

No state is “lost” or “created” as time evolves

$[p, 0, -H(q, p)]$

$$\vec{S} = -\nabla \times \vec{\theta}$$

(Minus sign to recover Ham eq)

KE

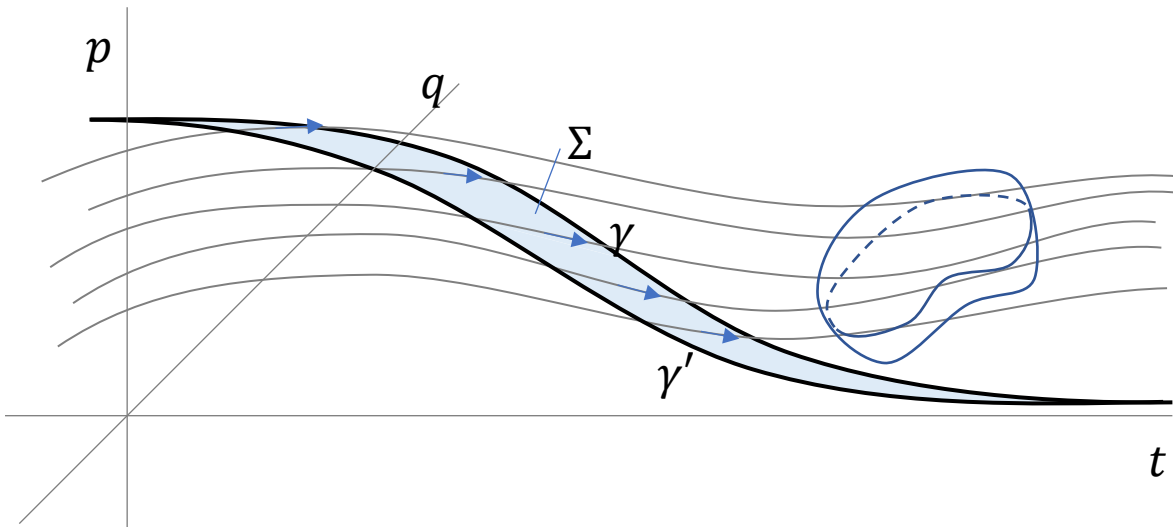
$p \frac{dq}{dt} + 0 \frac{dp}{dt} - H \frac{dt}{dt}$

$$\mathcal{A}[\gamma] = \int_{\gamma} L dt = \int_{\gamma} \vec{\theta} \cdot d\vec{\gamma}$$

Sci Rep **13**, 12138 (2023)

unphysical

The action is the line integral of the vector potential of the flow of states



Variation of the action

$$\begin{aligned} \delta \mathcal{A}[\gamma] &= \oint_{\partial \Sigma} \vec{\theta} \cdot d\vec{\gamma} \\ &= - \iint_{\Sigma} \vec{S} \cdot d\vec{\Sigma} \end{aligned}$$

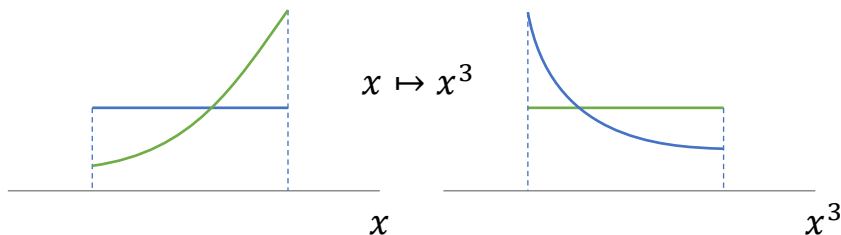
Gauge independent,
physical!

Variation of the action measures the flow of states (physical).

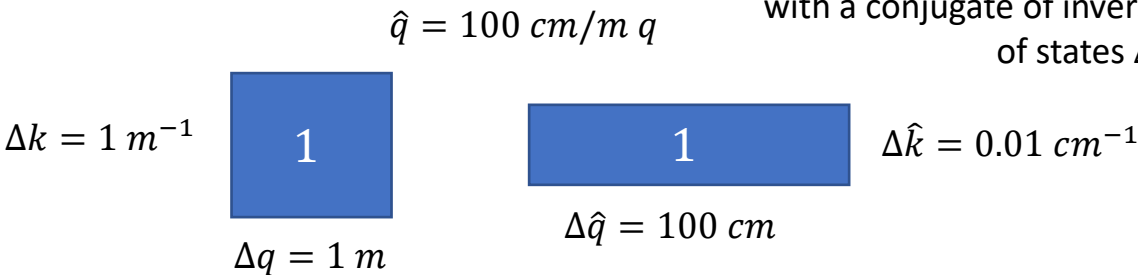
Variation = 0 $\Rightarrow$ flow of states tangent to the path.



Reversing phase-space



Density, entropy, uniform distributions
NOT in general coordinate invariant



Phase space (symplectic) structure is the only one that supports
coordinate invariant density, entropy, state count

- Phase space volumes are invariant under equal-time changes of coordinates q^i (PSV-VOL)
- The Jacobian for the transformation induced by equal-time changes of coordinates q^i is unitary (PSV-JAC)
- Densities over phase space are invariant under equal-time changes of coordinates q^i (PSV-DEN)
- Thermodynamic entropy is invariant under equal-time changes of coordinates q^i (PSV-THER)
- Information entropy is invariant under equal-time changes of coordinates q^i (PSV-INFO)
- Uncertainty of peaked distributions is invariant under equal-time changes of coordinates q^i (PSV-UNC)

- The system allows statistically independent distributions over each DOF under any choice of coordinates q^i (PSI-DEN)
- The system allows informationally independent distributions over each DOF under any choice of coordinates q^i (PSI-INFO)
- The system allows peaked distributions where the uncertainty is the product of the uncertainty on each DOF under any choice of coordinates q^i (PSI-UNC)

Invariance of entropy/state count + DOF independence
 $\Leftrightarrow$ structure of phase space (i.e. conjugate pairs)



Geometry is entropy

Classical mechanics

$$S(\rho_U) = \log \mu(U) = \log \int_U \omega$$

Symplectic form determines and is determined by the entropy of all uniform joint and marginal (DOF) distributions

$$\omega_{ab} = \begin{bmatrix} -mG_{\alpha\beta\gamma}u^\gamma - qF^{\alpha\beta} & mg_{\alpha\beta} \\ -mg_{\alpha\beta} & 0 \end{bmatrix}$$

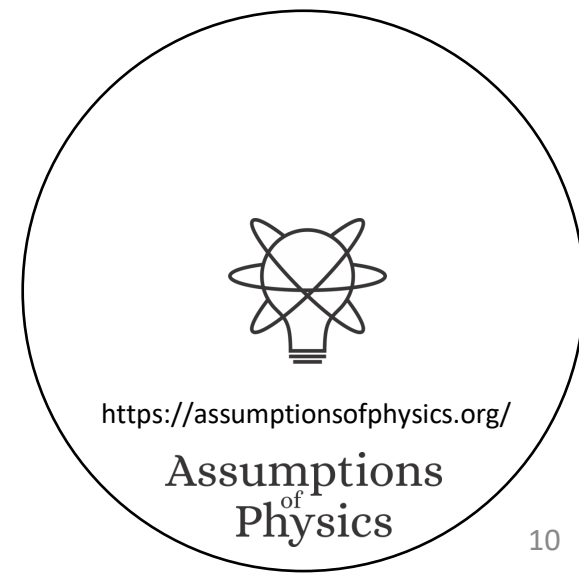
$\partial_\alpha g_{\beta\gamma} - \partial_\beta g_{\alpha\gamma}$

Metric tensor appears as the position-velocity part of the symplectic form

Quantum mechanics

$$S\left(\frac{1}{2}\rho_\psi + \frac{1}{2}\rho_\phi\right) = I\left(\frac{1+|\langle\psi|\phi\rangle|}{2}, \frac{1-|\langle\psi|\phi\rangle|}{2}\right) \\ = -\frac{1+|\langle\psi|\phi\rangle|}{2}\log\frac{1+|\langle\psi|\phi\rangle|}{2} - \frac{1-|\langle\psi|\phi\rangle|}{2}\log\frac{1-|\langle\psi|\phi\rangle|}{2}$$

Born rule determines and is determined by the entropy of all equal mixtures of pairs of pure states



Classical uncertainty principle

$$S(\rho) = -\int \rho \log h \rho dq dp$$

Uniform distribution over volume h has zero entropy

Fixes units

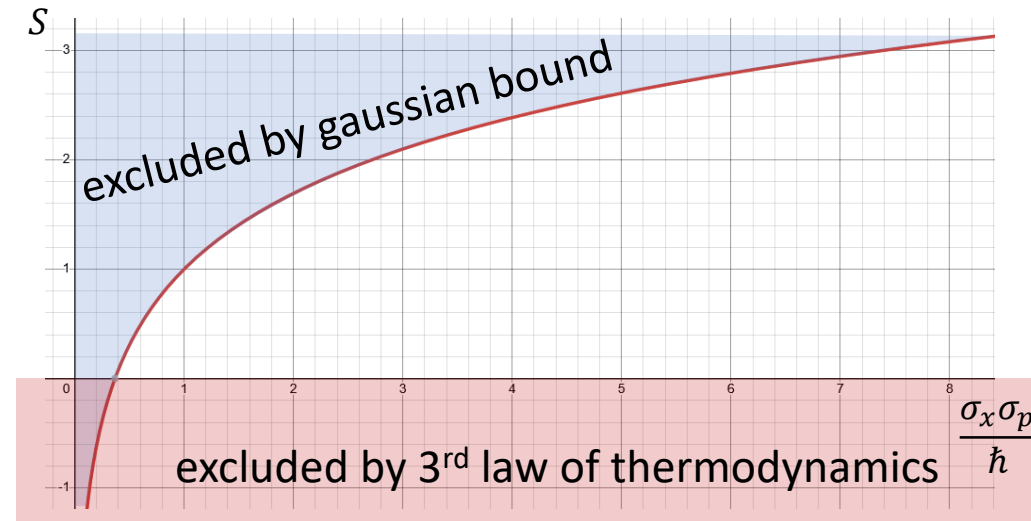
$$S(\rho) \leq \log 2\pi e \frac{\sigma_q \sigma_p}{h}$$

Gaussian maximizes entropy for a given uncertainty

$$\sigma_q \sigma_p \geq \frac{h}{2\pi e} e^{S(\rho)} = \frac{\hbar}{e} e^{S(\rho)}$$

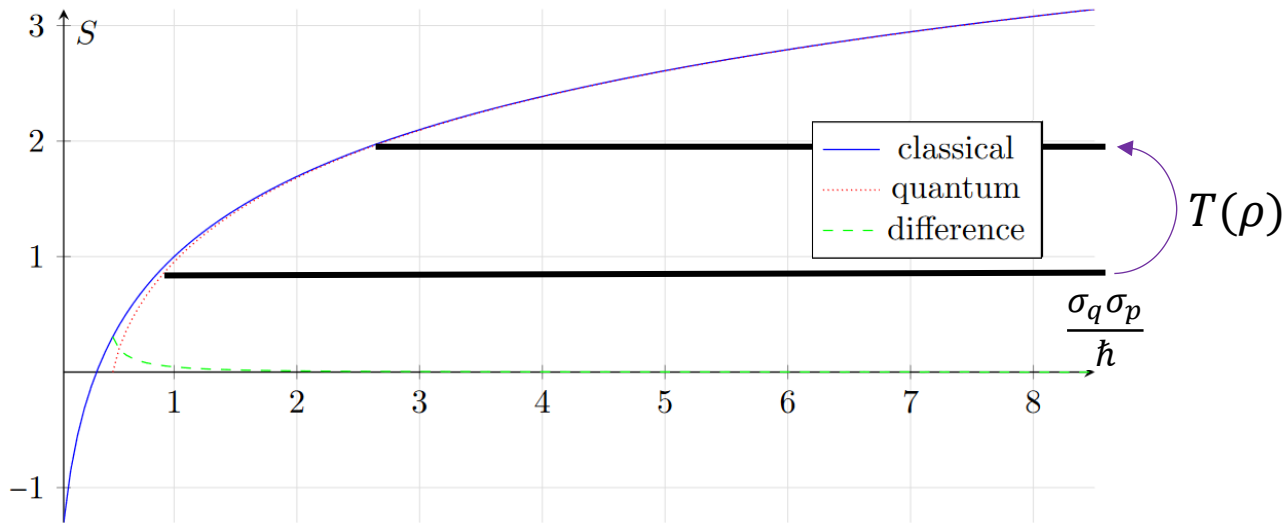
Entropy puts a lower bound on the uncertainty

$$S \geq 0 \Rightarrow \sigma_q \sigma_p \geq \frac{\hbar}{e}$$



Classical mechanics as high entropy limit

Physical limit: $S \gg 0$ like $\frac{v}{c} \ll 1$

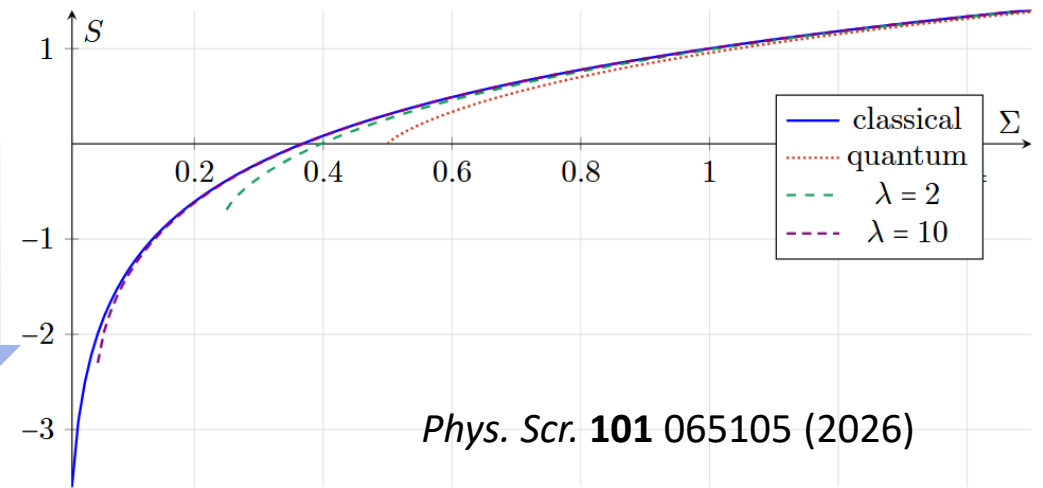


$$[X, P] = i\hbar \quad [T(X), T(P)] = \lambda i\hbar$$

Physical limit: increase the entropy of mixed states

Mathematical limit: decrease the entropy of pure states

Mathematical limit: $\hbar \rightarrow 0$ like $c \rightarrow \infty$



Phys. Scr. **101** 065105 (2026)

$$[T(\hat{X}), T(\hat{P})] = i\hbar$$

$$[\hat{X}, \hat{P}] = \frac{i\hbar}{\lambda}$$



Non-linear representation in QM

This map introduces a total phase based on all coefficients

$$m(c_+|z^+\rangle + c_-|z^-\rangle) = e^{i\theta \frac{|c_+|^2}{|c_+|^2 + |c_-|^2}} (c_+|z^+\rangle + c_-|z^-\rangle)$$

const. $\theta \in (0, 2\pi)$ Non-linear map...

... that preserves the physics
(i.e. Born rule)

$$\frac{\langle m(\psi) | m(\phi) \rangle \langle m(\phi) | m(\psi) \rangle}{\langle m(\psi) | m(\psi) \rangle \langle m(\phi) | m(\phi) \rangle} = \frac{\langle \psi | \phi \rangle \langle \phi | \psi \rangle}{\langle \psi | \psi \rangle \langle \phi | \phi \rangle}$$

Linearity is a mathematical convenience,
not a physical necessity



Superposition is multi-decomposition

Property of the **vector space representation**

$$\text{Superposition: } |\phi\rangle = c_1|\psi_1\rangle + c_2|\psi_2\rangle$$



Impossible in classical probability
(i.e. Choquet simplex)

$$\text{Multi-decomposition: } \rho = p_\phi\rho_\phi + p_{\hat{\phi}}\rho_{\hat{\phi}} = p_1\rho_{\psi_1} + p_2\rho_{\psi_2}$$

Linear transformations



Output of mixtures = mixture of outputs

Property of **distinguishability of preparations**

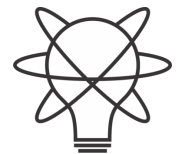
Indistinguishability of different mixtures



No physical process can depend on them



Paradoxes and standards of rigor



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Assumptions
of
Physics

Poor dimensional checking in statistical mechanics

Definition of Gibbs entropy

$$-\int_X \rho \log \rho \, dq^i \, dp_j \Rightarrow -\int_X \rho \log h^n \rho \, dq^i \, dp_j$$

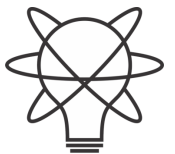
Density is not a pure number

When deriving Sackur–Tetrode equation

$$\Omega_N = \frac{1}{N!} \frac{V^N}{h^{3N}} \frac{2\pi^{3N/2}}{(\frac{3N}{2} - 1)!} (\sqrt{2mU})^{3N-1} \approx \frac{1}{N!} \frac{V^N}{h^{3N}} \frac{\pi^{3N/2}}{(3N/2)!} (\sqrt{2mU})^{3N}.$$

You can't “approximate physical dimensions”!!!

$$\text{length}^N \neq \text{length}^{N-1}$$



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Assumptions
of
Physics

Inconsistent expectations in Hilbert spaces

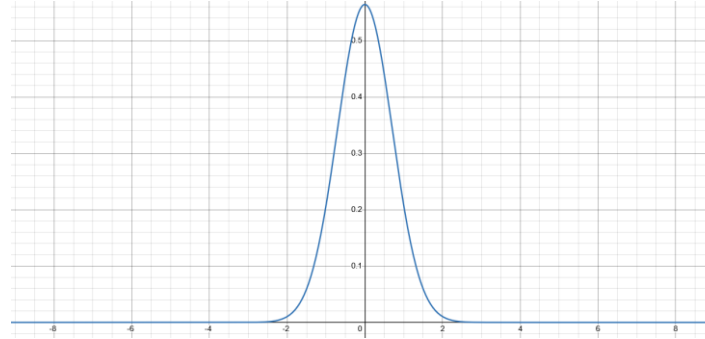
Quantum Stud.: Math. Found. **12**, 13 (2025)

$$\psi(x) = \sqrt{\frac{e^{-x^2}}{\sqrt{\pi}}}$$

$$\int |\psi|^2 dx = 1$$

$$\rho_\psi(x) = \frac{e^{-x^2}}{\sqrt{\pi}}$$

$$\langle X^2 \rangle_\psi = \frac{1}{2}$$

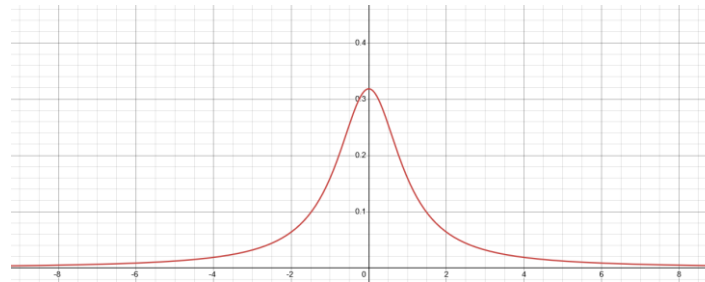


$$\phi(y) = \sqrt{\frac{1}{\pi(y^2 + 1)}}$$

$$\int |\phi|^2 dy = 1$$

$$\rho_\phi(y) = \frac{1}{\pi(y^2 + 1)}$$

$$\langle Y^2 \rangle_\phi \rightarrow \infty$$



Different observers see
finite/infinite expectation

$$y = \tan\left(\frac{\pi}{2} \operatorname{erf}(x)\right)$$

$$\psi(y) = \psi(x) \sqrt{\frac{dx}{dy}}$$

Expectation can have
finite/infinite oscillations

$$x(x_0, t) = x_0 \cos^2 \frac{\pi t}{2} + \tan\left(\frac{\pi}{2} \operatorname{erf}(x_0)\right) \sin^2 \frac{\pi t}{2}$$

Every continuous linear operator defined on the whole Hilbert space is bounded $\Rightarrow$ position/momentum/energy/number of particles are not defined on the whole Hilbert space!!!



Inconsistent convergence in Hilbert spaces

Consider the sequence of states given by

$$|\psi_i\rangle = \sqrt{\frac{i-1}{i}}|0\rangle + \sqrt{\frac{1}{i}}|i\rangle$$
$$\langle\psi_i|N|\psi_i\rangle = \frac{i-1}{i}\langle 0|N|0\rangle + \frac{1}{i}\langle i|N|i\rangle \rightarrow 1$$
$$|\psi_i\rangle \rightarrow |0\rangle$$

The state converges to the ground state
while the energy level converges to the first level



Inconsistent units for quantum observables

Suppose X is an observable

Expectation $\langle \psi | X | \psi \rangle$ should have units of X

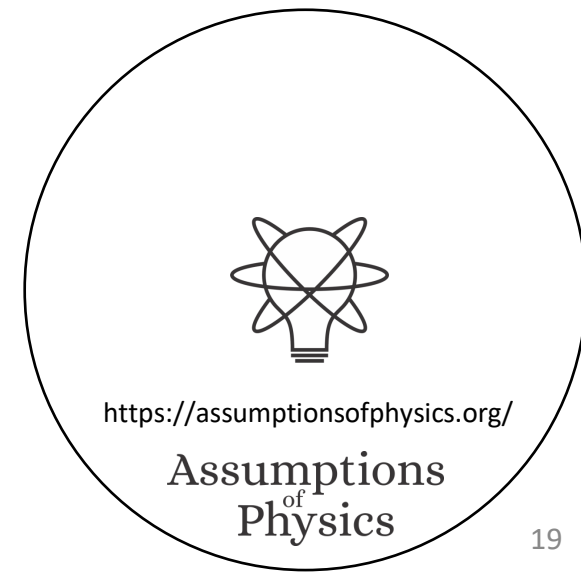
But then $X|\psi\rangle$ cannot have the same units of $|\psi\rangle$

But then $X|\psi\rangle + |\psi\rangle$ makes no physical sense

Therefore, X cannot be a map within the same space

Bonus: what are the units for $\psi(x)$?

Then why in some relativistic settings do they use the Fourier transform for a scalar function?



Tangent vectors in differential geometry

Defined as derivations

$$v: \mathcal{C}^\infty(X) \rightarrow \mathbb{R}$$

$$v = \sum_i v^i \partial_i$$

component basis

In polar coordinates

$$\partial_r + \partial_\theta = ???$$

[m] [rad]

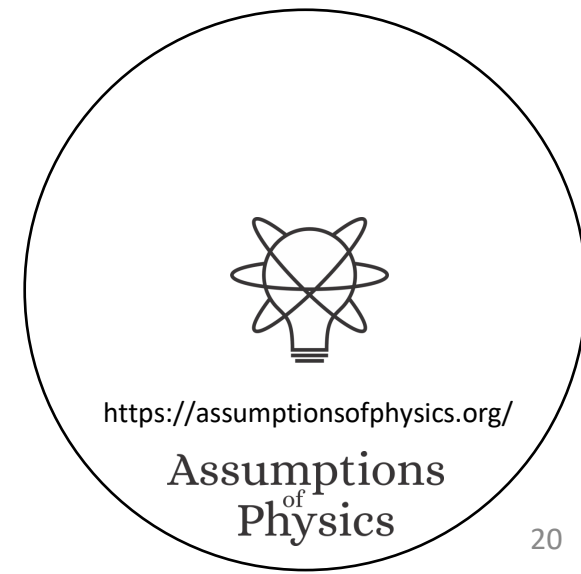
In phase space

$$\partial_q + \partial_p = ???$$

[m] [kg m s⁻¹]

Mathematically precise $\nRightarrow$ physically precise

Doesn't work with units



The Born rule is equivalent to the von Neumann entropy

Condition EQST (Ensembles and quantum states). *The space of ensembles $E(\mathcal{H})$ of a quantum system is represented by the convex space of all density operators (i.e positive semi-definite trace one self-adjoint operators) over a suitable separable inner product space $\mathcal{H}$. The state space is the set of extreme points, which coincides with the projective space $P(\mathcal{H})$. The ensemble space is equipped with a function $S : E(\mathcal{H}) \rightarrow \mathbb{R}$ where $S(\rho)$ returns the entropy for ensemble ρ . The ensemble space is also equipped with a function $p(\cdot|\cdot) : P(\mathcal{H}) \times E(\mathcal{H}) \rightarrow [0, 1]$, where $p(\psi|\rho)$ returns the probability of finding ψ having prepared ρ . This function is affine in the second argument (i.e. $p(\phi|\sum p_i \psi_i) = \sum_i p_i p(\phi|\psi_i)$) and returns one if and only if the initial and final states are the same (i.e. $p(\phi|\psi) = 1 \iff \phi = \psi$ for all $\psi, \phi \in P(\mathcal{H})$).*

Condition BR-BORN (Born rule). *The conditional probability p is given by the Born rule:*
$$p(\psi|\phi) = \frac{\langle \phi|\psi \rangle \langle \psi|\phi \rangle}{\langle \psi|\psi \rangle \langle \phi|\phi \rangle}.$$

Condition BR-ENT (Von Neumann entropy). *The entropy function S is given by the Shannon entropy: $S(\rho) = -\text{tr}(\rho \log \rho)$.*

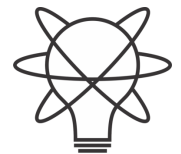
Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in P(\mathcal{H})$ be a countable sequence of states. Then $p(\psi_i|\psi_j) = 0$ for all $i \neq j$ if and only if $S(\sum_i p_i \psi_i) = -\sum_i p_i \log p_i$ for all p_i such that $\sum_i p_i = 1$.*

Theorem 9. *Over **EQST**, if two conditions among **BR-BORN**, **BR-ENT** and **[B]MUTEX** are assumed, the third is implied.*

We CAN analyze logical dependencies between conditions that are both mathematically precise and physically meaningful



Logical dependencies and base theories



<https://assumptionsofphysics.org/>

Assumptions
of
Physics

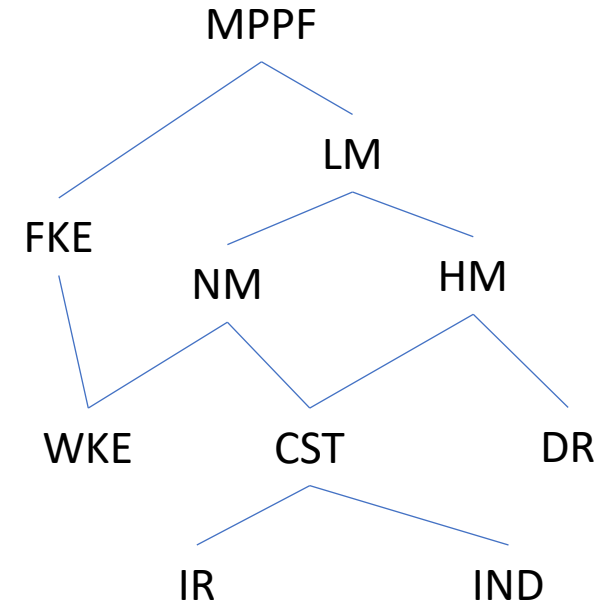
Classical conditions and their relationships

Core assumptions

IR: Infinitesimal Reducibility
 IND: INDependent DOF
 W/FKE: Weak/Full Kinematic Equivalence
 DR: Determinism+Reversibility

Physical theories

CST: Classical States
 NM: Newtonian Mechanics
 HM: Hamiltonian Mechanics
 LM: Lagrangian Mechanics
 MPPF: Massive Particles under Potential Forces



Each assumption is an equivalence class of multiple conditions

There is a linear relationship between conjugate momentum and velocity

(FKE-LIN)

At each position, there exists a local inertial frame

(FKE-INER)

The position fully defines the units of all state variables, and therefore an invertible transformation between momentum and velocity

(FKE-UNIT)

Density expressed in velocity at the same position is proportional to the density over states

(FKE-PROP)

Uniform distributions along momentum correspond to uniform distributions along velocity

(FKE-UNIF)

The system under study is a massive particle under scalar and vector potential forces

(FKE-POT)

The Jacobian of the transformation between state variables and kinematic variables is a non-singular function of position only.

(FKE-NSIN)

Densities over phase space can be expressed in terms of position and velocity by rescaling the value at each point: $\rho(x^i, v^j) |J(x^i)| = \rho(q^i, p_j)$.

(FKE-DEN)

Areas and volumes in phase space can be expressed in kinematic variables, and the transformation depends on position only: $dx^1 \dots dx^n dv^1 \dots dv^n = |J(x^i)| dq^1 \dots dq^n dp_1 \dots dp_n$.

(FKE-VOL)

The symplectic form ω_{ab} can be expressed in kinematic variables, and its components are a linear function of velocity.

(FKE-SYMP)



Quantum pure states are not enough

Condition QP-RAY (Quantum states are rays). *The state space of the system is represented by the projective space given by the rays of a complex vector space $\mathcal{H}$. That is, a state $\psi \in P(\mathcal{H})$ is a ray $\psi = \{c|\psi\rangle \mid c \in \mathbb{C}\}$ of a complex projective space.*

Rays do not need the inner product

Condition CST (Classical state space). *The state space is represented by a symplectic manifold (X, ω) , typically charted by conjugate quantities (i.e. local Darboux coordinates) $(q_i, p_i) \in \mathbb{R}^{2n}$, where n is the number of independent degrees of freedom (DOF). The count of configurations for each DOF over a region Σ is given $\int_{\Sigma} dp^i dq_i = \int_{\Sigma} \omega$ and the count of states is given by the Liouville measure $\mu(U) = \int_U dq^1 \cdots dq^n dp_1 \cdots dp_n = \int_U \omega^n$.*

The ensemble space is represented by the set of probability distributions $\rho(q^i, p_i)$ (i.e. probability measures that are absolutely continuous with respect to the Liouville measure).

The conditional probability over states is given by $p(\xi_1|\xi_2) = 0 = p(\xi_2|\xi_1)$ for all $\xi_1, \xi_2 \in X$, meaning that all states represent mutually exclusive events.

Theorem 1. *Conditions QP-RAY and CST are consistent. In fact, they are consistent if and only if $1 \leq \dim(\mathcal{H}) < \infty$.*

You CAN endow quantum pure states with classical probability distributions and Hamiltonian mechanics

E.g. Classical Larmor precession



Quantum ensembles rule out classical mechanics

Condition QE-DENS (Quantum ensembles are density operators). *The ensemble space of the system is represented by the set of density operators acting on a complex inner product space $\mathcal{H}$. That is, an ensemble $\rho \in D(\mathcal{H})$ is a positive semi-definite trace one self-adjoint operator $\rho: \mathcal{H} \rightarrow \mathcal{H}$.*

Condition BR-BORN (Conditional expectation from Born rule). *The conditional probability $p: P(\mathcal{H}) \times P(\mathcal{H}) \rightarrow [0, 1]$ is given by the Born rule. That is, $p(\psi|\phi) = \frac{\langle \phi|\psi \rangle \langle \psi|\phi \rangle}{\langle \psi|\psi \rangle \langle \phi|\phi \rangle}$.*

Trace requires
the inner product!

Theorem 3. *Conditions QE-DENS and CST are inconsistent.*

Classical and quantum ensembles are incompatible

Since BR-BORN is independent from QE-DENS,
QE-DENS is responsible for the incompatibility



Base theory for all of physics

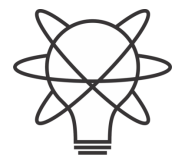
Collection of conditions that MUST be true in all physical theories

Condition EQST (Ensembles and quantum states). *The space of ensembles $E(\mathcal{H})$ of a quantum system is represented by the convex space of all density operators (i.e. positive semi-definite trace one self-adjoint operators) over a suitable separable inner product space $\mathcal{H}$. The state space is the set of extreme points, which coincides with the projective space $P(\mathcal{H})$. The ensemble space is equipped with a function $S : E(\mathcal{H}) \rightarrow \mathbb{R}$ where $S(\rho)$ returns the entropy for ensemble ρ . The ensemble space is also equipped with a function $p(\cdot|\cdot) : P(\mathcal{H}) \times E(\mathcal{H}) \rightarrow [0, 1]$, where $p(\psi|\rho)$ returns the probability of finding ψ having prepared ρ . This function is affine in the second argument (i.e. $p(\phi|\sum p_i \psi_i) = \sum p_i p(\phi|\psi_i)$) and returns one if and only if the initial and final states are the same (i.e. $p(\phi|\psi) = 1 \iff \phi = \psi$ for all $\psi, \phi \in P(\mathcal{H})$).*

Condition [B]MUTEX (Mutual exclusivity conditions). *Let $\psi_i \in P(\mathcal{H})$ be a countable sequence of states. Then $p(\psi_i|\psi_j) = 0$ for all $i \neq j$ if and only if $S(\sum p_i \psi_i) = -\sum p_i \log p_i$ for all p_i such that $\sum p_i = 1$.*

Part of the base theory

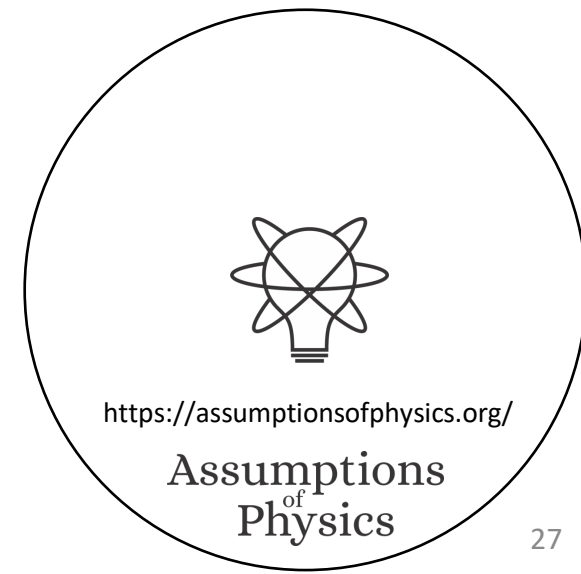
Base theory $\neq$ Theory of everything



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Assumptions
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Physics

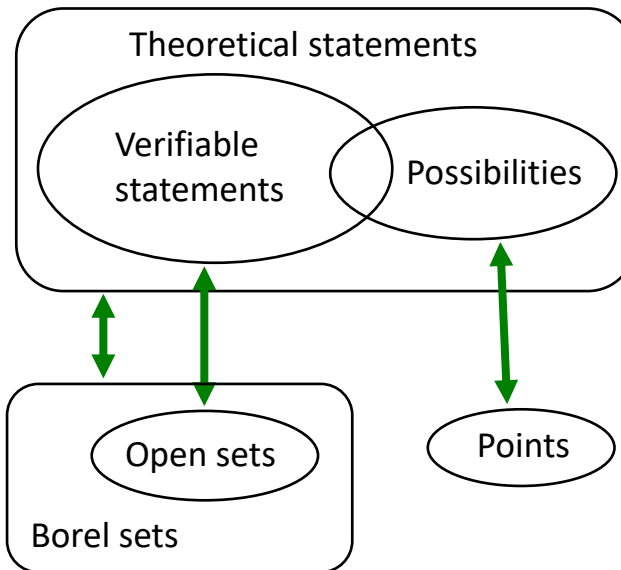
Foundational pillars



Experimental verifiability

Science is evidence based $\Rightarrow$ Statements are associated to experimental tests

Verifiable statement	Test Result
T	SUCCESS (in finite time)
F	UNDEFINED FAILURE (in finite time)



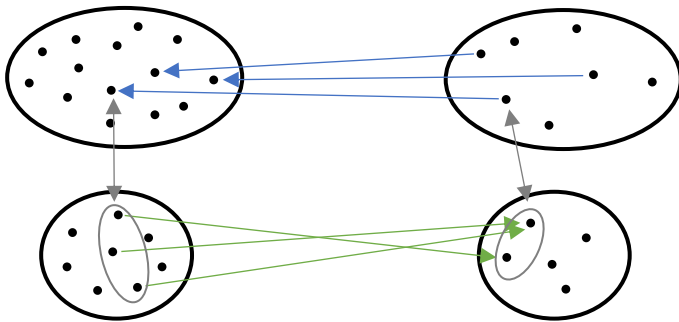
s	Test
T	S
F	U
	U
	F

$int(A)$ corresponds to the verifiable part of a statement

∂A corresponds to the undecidable part of a statement

$ext(A)$ corresponds to the falsifiable part of a statement

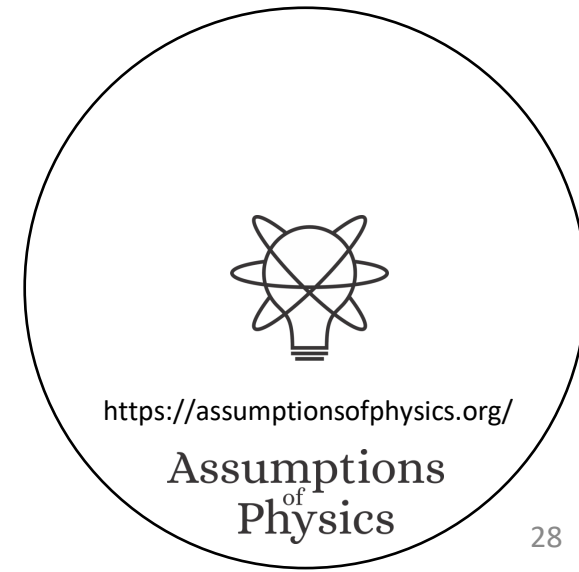
Inference relationship $r: \mathcal{D}_Y \rightarrow \mathcal{D}_X$ such that $r(s) \equiv s$



Causal relationship $f: X \rightarrow Y$ such that $x \preceq f(x)$

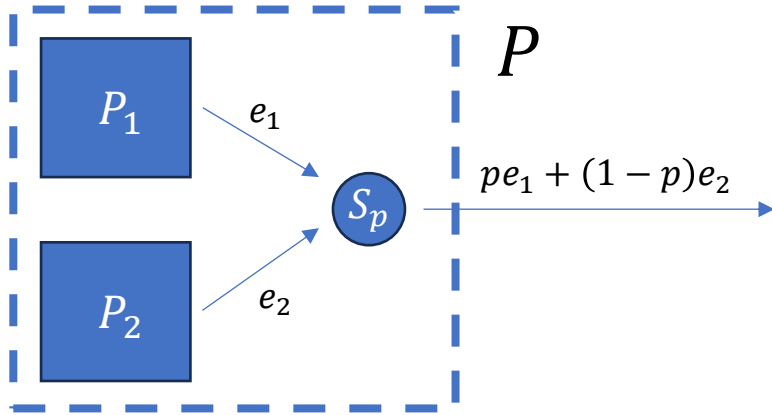
**Experimental verifiability
 $\Rightarrow$ topology and σ -algebra**

Experimental relationships must be topologically continuous



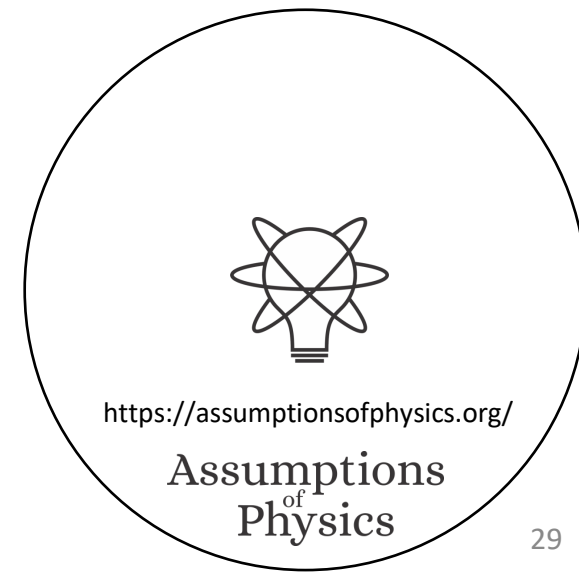
Reproducibility and ensembles

Scientific laws describe reproducible relationships $\Rightarrow$ Ensembles



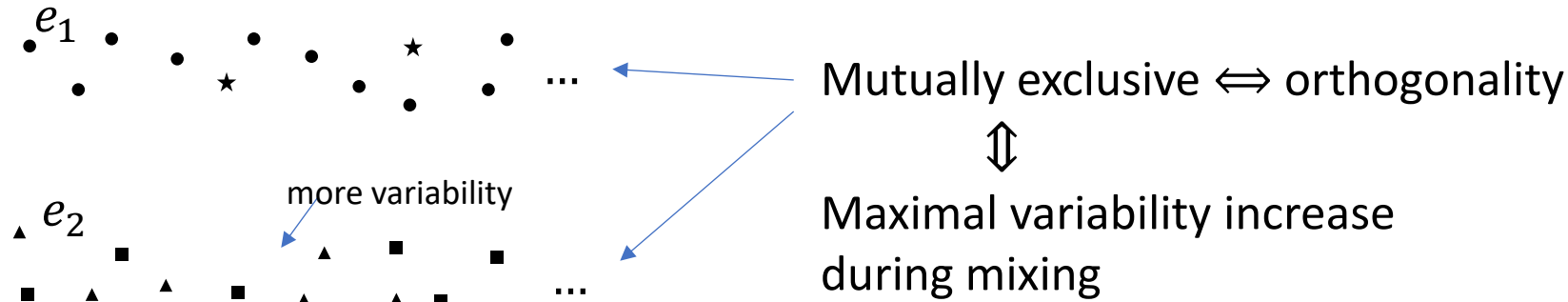
Ensembles are outputs of repeatable processes
Statistical mixing $\Rightarrow$ Linear structures in physics

Note: mixing coefficients are NOT probabilities



Ensemble variability

Variability of the elements within an ensemble $\Rightarrow$ Entropy

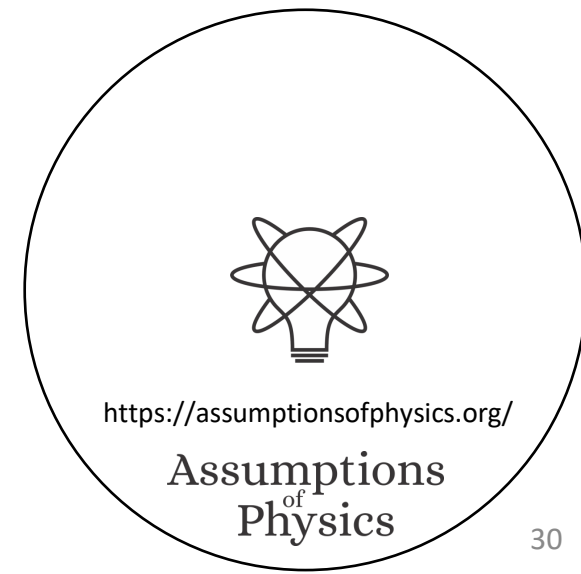


Variability $\Rightarrow$ Probabilities, inner product and geometry

Probability $\Leftrightarrow$ Mixtures of mutually exclusive ensembles

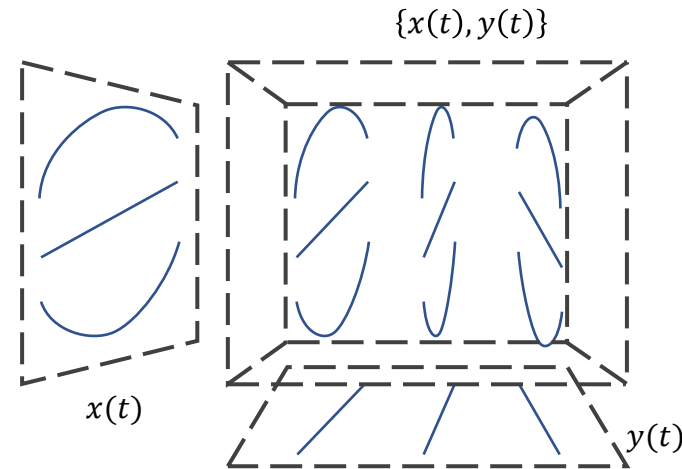
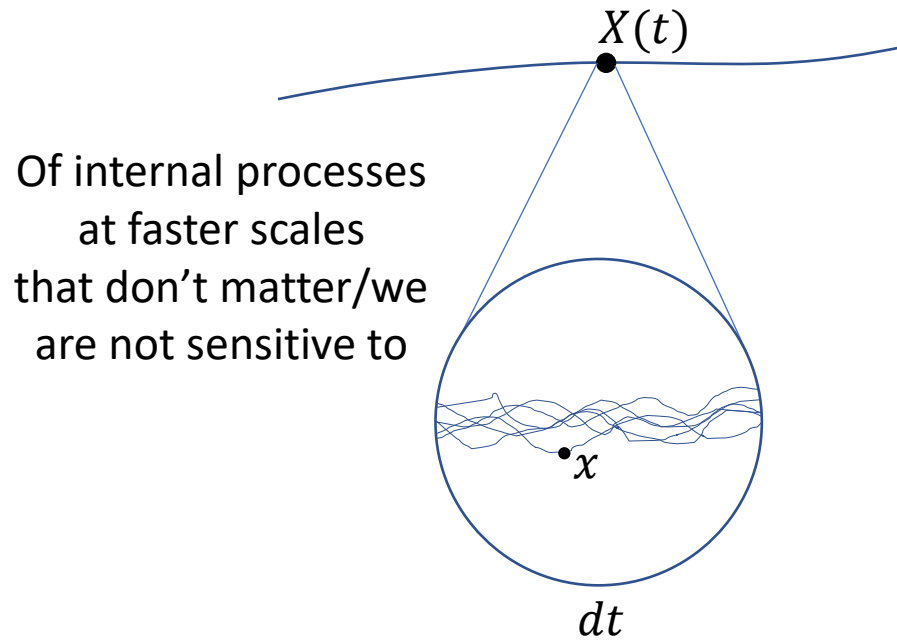
Classical mechanics/probability $\Leftrightarrow$ All pure states are mutually exclusive

Pure states $\Leftrightarrow$ Best preparations $\nRightarrow$ No variability



Equilibria

States must exist for at least a short time $\Rightarrow$ Equilibria

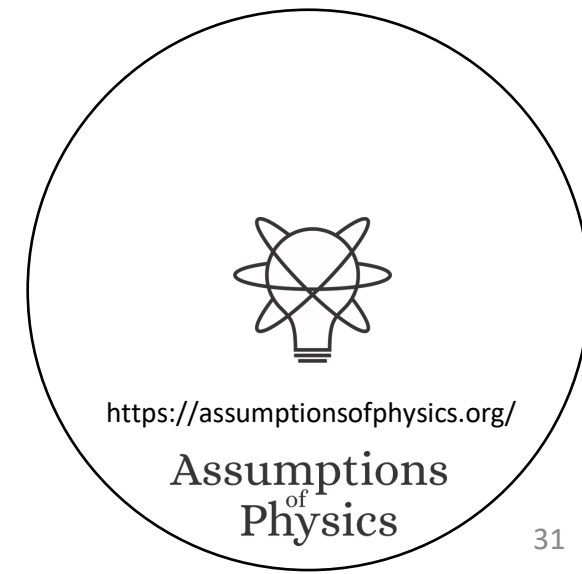


Of system/environment
interactions
that don't matter/we
are not sensitive to

Least developed part

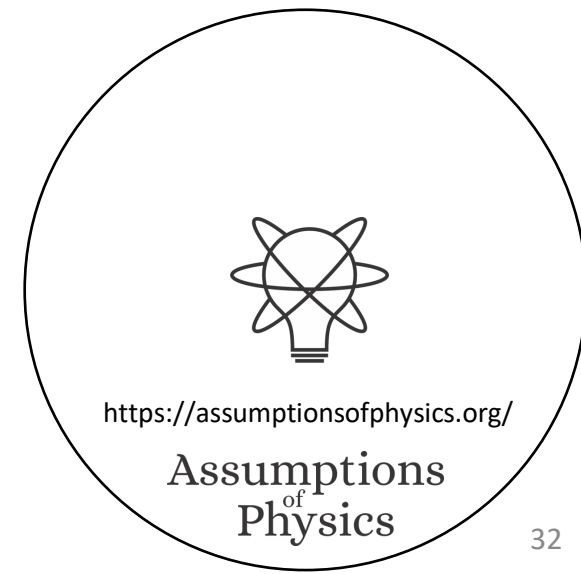
This requirement allows us to properly talk about different physical objects

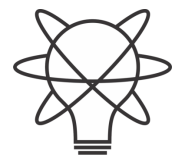
Equilibria $\Rightarrow$ Symmetries, differentiability, Lie algebras



The foundations of physics has a methodology problem

Reverse Physics gives us a powerful methodology, more in line with foundational approaches of other disciplines, that gives us a lot more and more precise insights and opens new avenues of research





<https://assumptionsofphysics.org/>

**Assumptions
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Physics**